

CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

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Membrane Builder

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PDB Info	STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6	JOB ID: 3750189754
Oriented PDB: step2_orient.pdb (view structure)							download.tgz
System Size Input: step3_size.inp							
System Size Output: step3_size.out							
System Size: step3_size.str							
Packing Simulation: step3_packing.inp							Packing Simulation Input
step3_packing.out							Packing Simulation Output
crystal_image.str							Crystal Image
step3_packing_pol.str							Topology File of Pseudo Lipid Spheres
step3_packing.pdb (view structure)							Generated Packed System (please view this structure before you move to the next step)

Determined System Size:

Box Type	Rectangle		
Crystal Type	TETRAGONAL		
System Size	A	94.9137157	Dimension along the A (X) axis
	B	94.9137157	Dimension along the B (Y) axis
	C	94.2717514	Dimension along the C (Z) axis
Crystal Angle	Alpha	90.0	Angle between the axis B and C
	Beta	90.0	Angle between the axis A and C
	Gamma	90.0	Angle between the axis A and B
# of Lipids	on Top	115	
	on Bottom	112	
Z Center	-0.4482599		Center of the system along the Z axis

System Building Options:

- ☐ Insertion method Build system using insertion method
☒ Replacement method Build system using replacement method
☒ Check lipid ring (and protein surface) penetration

For this system, insertion method can not be used. Replacement method will be used instead.

Component Building Options:

☒ Include IonsIon Placing Method: Distance ☒ Basic Ion TypesKCl ☐ More Ion Types**Formula Cation Anion Concentration Neutralizing**

KCl

☒

Ion	Count
K ⁺	39
Cl ⁻	36

Please note that the ion count is an approximation based on geometry. The real number will be calculated in the next step.

Pore Water Options:

The addition of pore water may result in water molecules placed outside the protein. Please download the structure file and open it using your molecular visualization software. In the list below, click water molecules that are outside the pore and should be removed.

Water filled protein structure: [step2.1_pore.pdb](#)

Check atom number of OH2 to remove

	1	2	3	4	5	6	7	8	9	10		1	2	3	4	5	6	7	8	9	10		
1	5401	5404	5407	5410	5413	6	5416	5419	5422	5425	5428	11	5431	5434	5437	5440	5443	16	5446	5449	5452	5455	5458
21	5461	5464	5467	5470	5473	26	5476	5479	5482	5485	5488	31	5491	5494	5497	5500	5503	36	5506	5509	5512	5515	5518
41	5521	5524	5527	5530	5533	46	5536	5539	5542	5545	5548	51	5551	5554	5557	5560	5563	56	5566	5569	5572	5575	5578
61	5581	5584	5587	5590	5593	66	5596	5599	5602	5605	5608	71	5611	5614	5617	5620	5623	76	5626	5629	5632	5635	5638
81	5641	5644	5647																				

Next Step: 
Build Components



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